Enhancing Physical Consistency in Lightweight World Models

Dingrui Wang, Zhexiao Sun, et al.

THE PROBLEM: ROBOTS HAVE ”FEVER DREAMS”

Imagine you are driving a car. Before you turn the steering wheel, you quickly play a movie in your head: *”If I turn left, will I hit that truck?”* This ability to simulate the future is called a **World Model**.

Big AI models (like Sora or Genie) are great at this. They understand physics in a good manner. But they are enormous. Running them requires a massive supercomputer.

If you try to run these ”dreams” on a regular robot chip (like the one in a Tesla or a small drone), you have to use a lightweight, compressed model. The problem? **Lightweight models get the physics wrong.** Cars in the simulation start disappearing, teleporting, or merging into each other. It is like the robot is having a fever dream—it looks trippy, but you definitely should not trust it to drive your car.

Our Goal: We want to make a lightweight AI model that understands physics just as well as the giant ones, so it can run on a robot's actual onboard computer.

THE SOLUTION: SIMPLE TRICKS

We did not just make the model bigger. Instead, we taught it to focus. We used ”human-like” intuitions to fix the physics:

The ”Highlighter” Trick (Soft Mask)

Metaphor: Imagine reading a textbook. If you highlight the key words with a neon pen, you remember them better.

Existing methods either give the AI the raw image (which is too messy) or a strict black-and-white mask of objects (which is too rigid, like blocking out the text).

We use a **”Soft Mask.”** Think of it as a gentle glow wrapping around every car on the road, including the ego vehicle and the surrounding vehicles. We feed this ”glow” into the AI during training and inference. It gently tells the AI: *”Hey, pay attention to this area, there is a moving object here. Keep its motion flexible and responsive to actions.”* This forces the small brain to focus its limited processing power on what matters: collisions and movement.

THE RESULT: PHYSICALLY SMARTER, EDGE-READY

We tested this in a traffic simulator. Here is the punchline:

- **It is Physically Smarter:** At the same 400M parameter size as the standard baseline, our physics-informed world model achieves about a **60% boost** in overall physical-consistency scores.
- **It Stays Strong When Smaller:** Even a 130M-parameter version still beats the 400M baseline, with roughly **7.4% higher** physical-consistency scores and about **28% faster** inference, making it suitable for edge deployment.
- **It Works in Practice:** These improvements come with essentially the same compute footprint as the baseline, and the model runs fast enough on conventional edge computing devices.